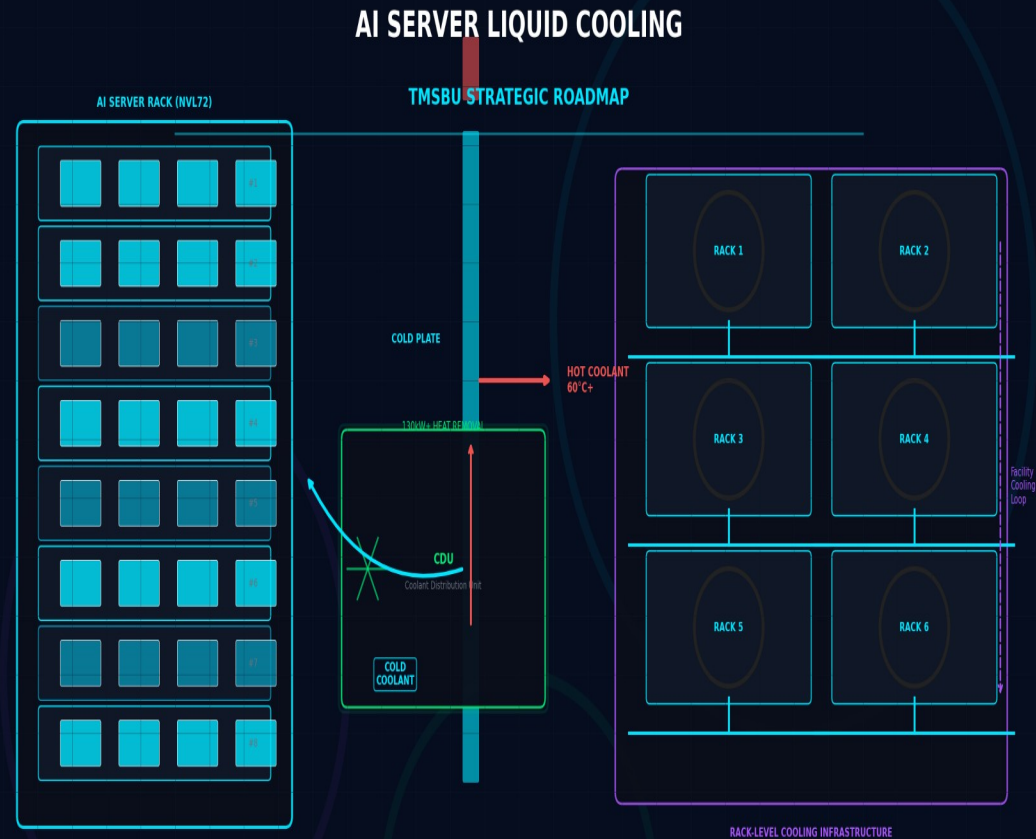


AI Server Liquid Cooling Strategic Roadmap

TMSBU Commercial Opportunity
Analysis



Executive Summary

02

液冷是商業模式轉型，不是產品升級

TREND

1,400W

GPU TDP @ 2027+

MARKET

60%+

new AI DC by 2026

HYBRID

BOTH

TMSBU 風冷液冷皆受益

OPPORTUNITY

\$1.8B

SOM (需內部驗證)

ACTION

NOW

6-18M 關鍵窗口

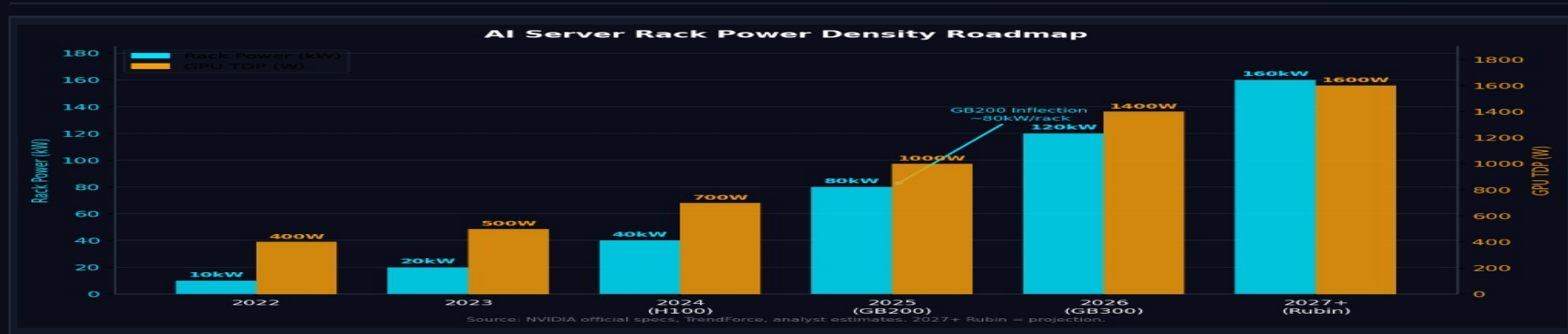
KEY INSIGHTS

- 散熱瓶頸是 AI 基礎建設擴張的硬約束 — 誰能解決、誰就能贏得市佔
- 液冷市場 TAM \$39.1B (CAGR 45%)，現在是佈局關鍵時刻
- Delta 集團唯一同時具備風冷 + 液冷 + Power+Control 全段整合能力

Thermal Bottleneck

03

GPU TDP 突破風冷散熱上限，散熱瓶頸成為 AI 基礎建設擴張的硬約束



1,400W

GPU TDP @ 2027+

160kW

Rack Power @ 2027+

\$39.1B

Liquid Cooling TAM by 2030

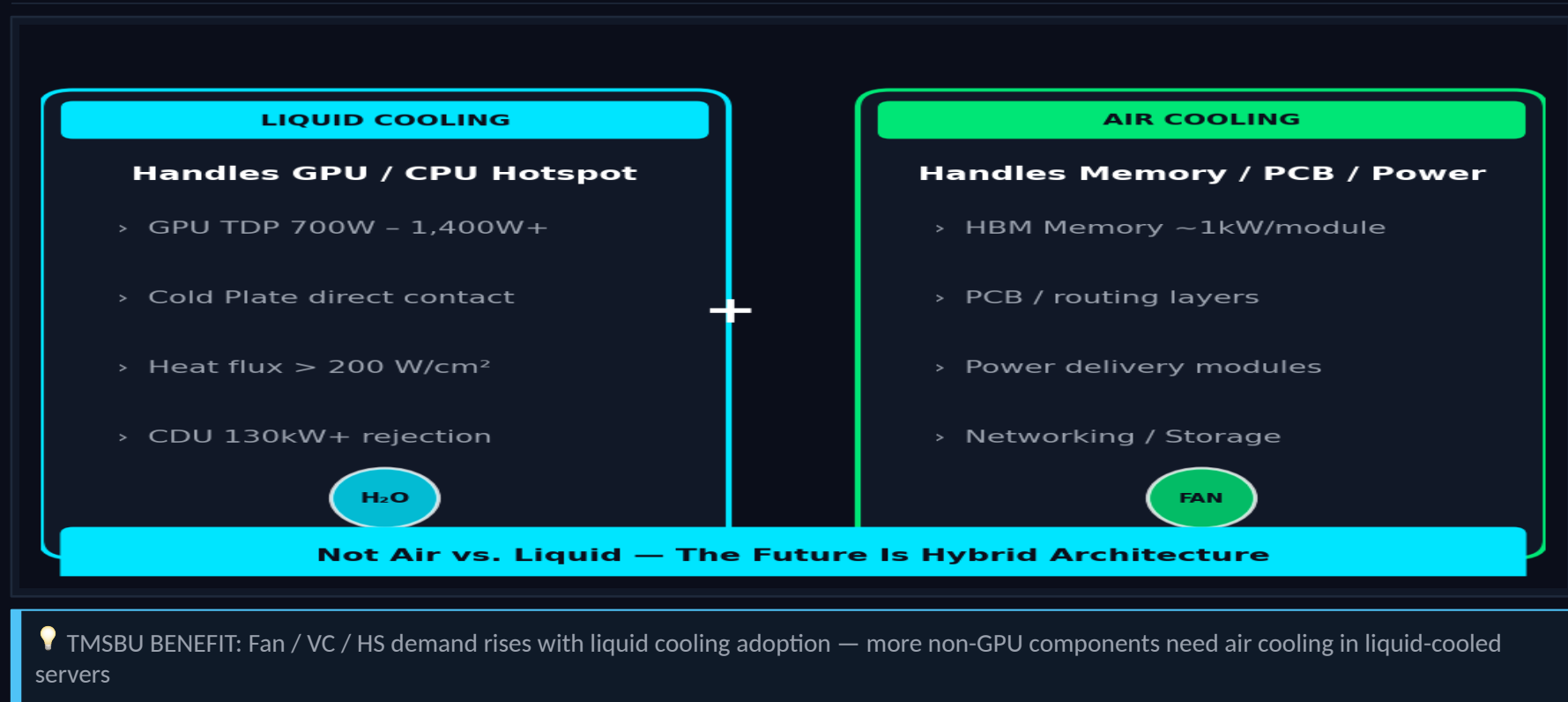
KEY INSIGHTS

- GB200 NVL72 (80-130kW/rack) 是風冷物理瓶頸的轉捩點
- Rubin (2027+) 功率密度挑戰 160kW+，浸沒式冷卻可能是下一個轉折
- 散熱瓶頸等於誰能解決、誰就能贏得 AI 基礎建設市佔

Why Air Cooling Alone Is No Longer Enough

04

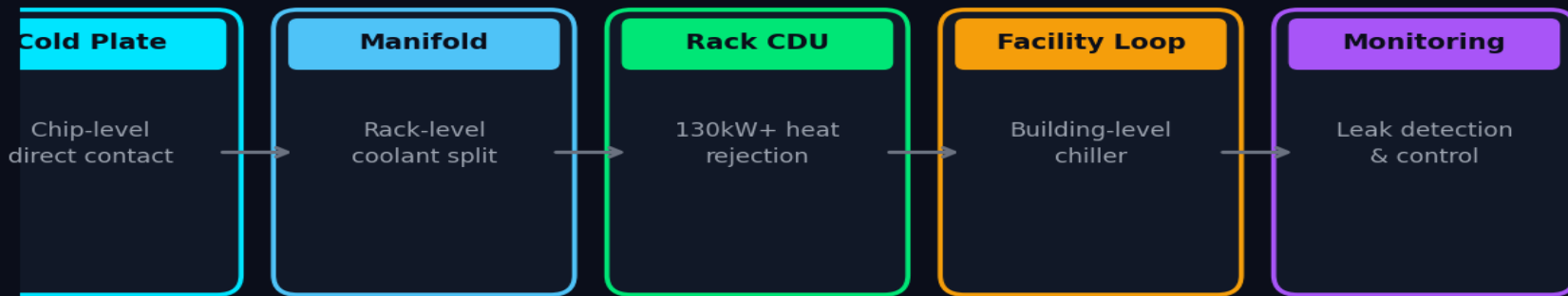
Not Air vs. Liquid — The Future Is Hybrid Architecture



Standard

05

液冷的價值在完整系統整合：Cold Plate → CDU → Manifold → Quick Connector



System integration complexity = highest value source · Vendor complete = highest margin

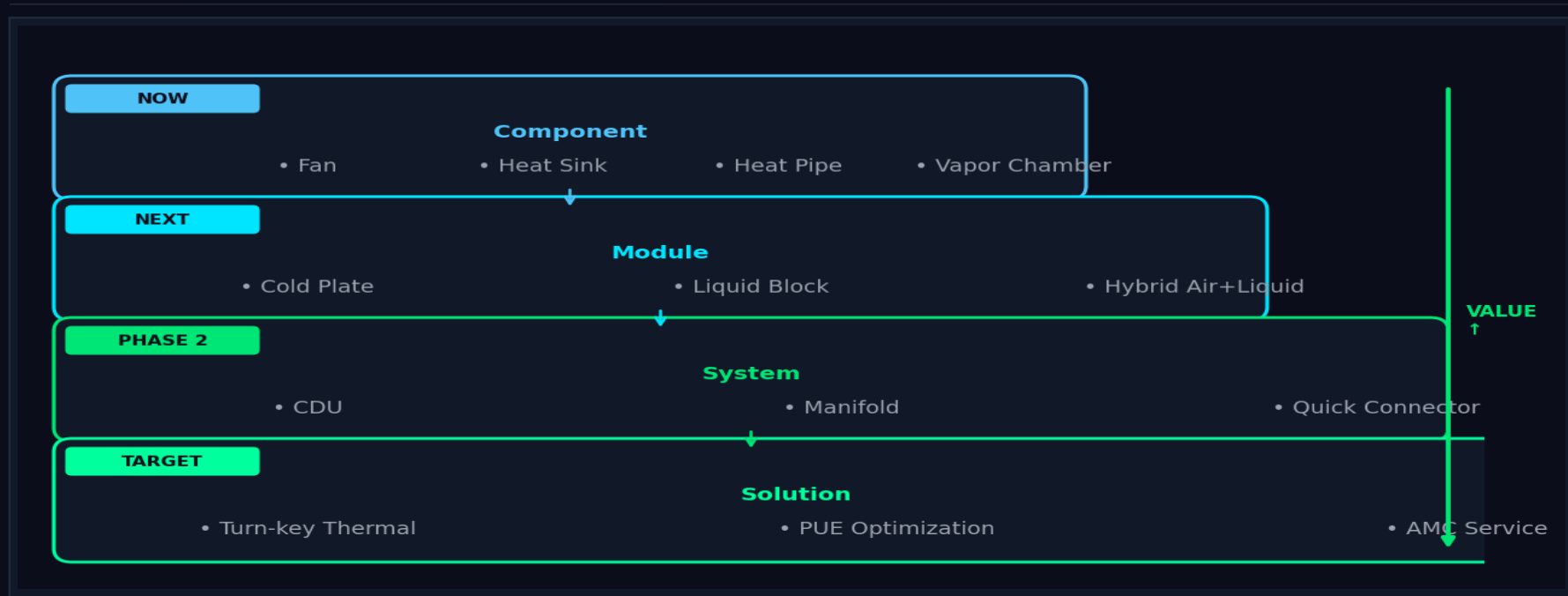
KEY INSIGHTS

- Cold Plate + CDU + Manifold + Quick Connector + Leak Detection = 完整液冷系統
- 系統整合複雜度是最高附加價值來源 — 不是單一零件
- 能提供完整系統的 Vendor，才能拿下最高市佔與毛利

System Solution

06

自然擴張路徑：零件 → 模組 → 系統 → 解決方案



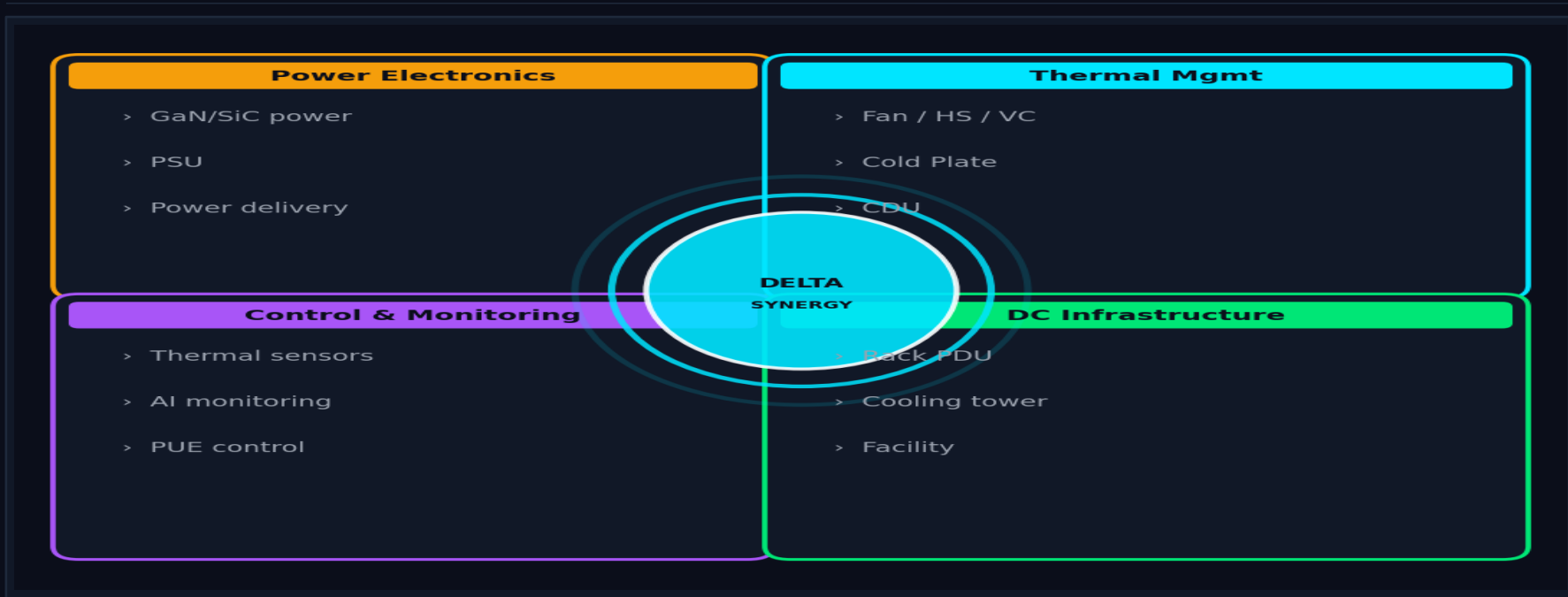
KEY INSIGHTS

- CDU + Quick Connector + Manifold = TMSBU 最高價值的擴張方向
- 從零件報價升級到系統方案統包報價，改變商業模式
- 目標：成為客戶不可或缺的 Thermal System Partner

Cooling

07

唯一同時具備風冷與液冷能力的供應商 + Delta 集團層級綜效



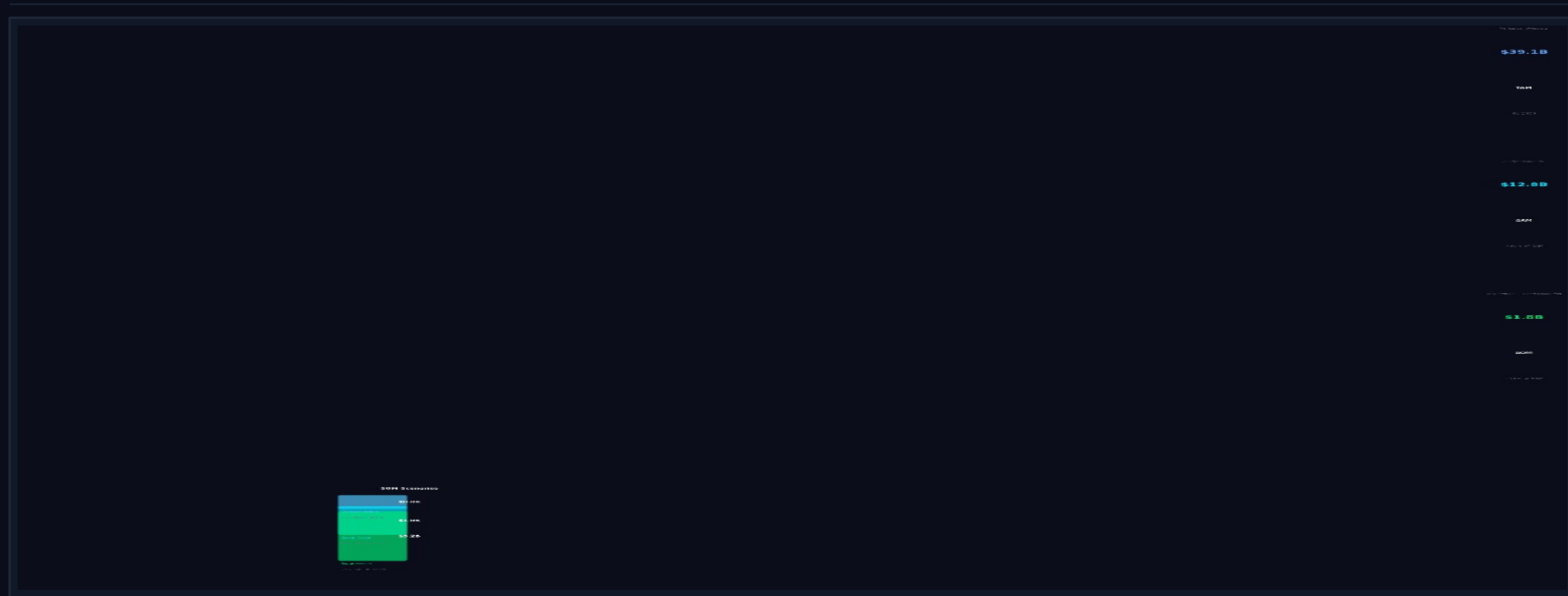
KEY INSIGHTS

- Delta 集團唯一能 Chip-level → Rack-level → Facility-level 全段熱管理整合
- Fan + Power + Control + DC Infra 集團綜效是競爭對手無法快速複製的護城河
- Manufacturing scale + Global support network 支撐 Tier-1 客戶信任

Market Opportunity: TAM / SAM / SOM

08

系統層級解決方案擴大 TMSBU 可觸及市場規模



KEY INSIGHTS

- TAM \$39.1B (M&M 2024) 涵蓋整個 AI 資料中心液冷市場
- SOM = 假設 TMSBU 滲透率 14% of SAM , 需要 James 內部驗證
- 三情境：保守 \$0.8B / 基準 \$1.8B / 積極 \$3.2B

△ SOM = assumption, needs internal TMSBU validation. TAM: MarketsandMarkets. SAM: analyst estimate. Scenarios: TMSBU analysis.

Capability Gap & Risk Assessment

09

3 項最高優先風險：CDU 能力 / 洩漏可靠性 / 客戶認證週期

Risk Item	Impact	Urgency
CDU Capability	HIGH	HIGH
Leak Reliability	HIGH	HIGH
Customer Qual Cycle	HIGH	HIGH
Quick Connector Qual	HIGH	MED
Coolant Compatibility	MED	MED
Thermal Simulation	MED	MED
Integration Resp.	MED	LOW
Global Support	LOW	LOW

① CDU Strategy

Partner / Acquire / In-house
Decision within 90 days

② Leak Reliability

Quick connector qual
Leak detection mandatory

③ Customer Qual

Start NOW
12-18 month cycles

KEY INSIGHTS

- CDU Strategy 是 Phase 1 最關鍵決策 — 合作 / 併購 / 自製，90 天內鎖定
- 洩漏可靠性是紅線：單次洩漏可摧毀 \$50 萬 server，不可妥協
- 客戶認證週期 12-18 個月，現在啟動才能趕上 GB300/Rubin

Strategic Roadmap & Recommended Actions

10

4 階段 6-18 個月 Roadmap : Phase 1 基礎 → Phase 4 規模化



KEY INSIGHTS

- Phase 1 最關鍵決策：CDU 是內部開發 / 合作夥伴 / 併購？這將決定 TMSBU 液冷成敗
- 所有 Phase 終極目標：從 Component Supplier 轉型為 Thermal System Solution Provider
- 落後代價：失去 GB300/Rubin 平台供應商資格，影響未來 5-10 年競爭地位